PROJECT BRIEF

Child Survival 2030

Which countries will meet the 2030 targets for child and maternal survival, and how much faster do the others need to move?

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1. Purpose

This project builds an interactive Power BI dashboard on child and maternal deaths in every country from 1990 to the latest estimates, measured against the Sustainable Development Goal (SDG) targets for 2030. It is an exploratory analysis and visualisation project: no machine learning, but real analytical work, including rates, population weighting, rates of reduction and simple trend projections.

It follows two earlier portfolio dashboards (a synthetic cardiovascular risk dashboard and a food price dashboard for North East Nigeria). The aim here is a step up in both the question asked and the Power BI techniques used.

2. Why it matters

In 2024, about **4.9 million children** died before their fifth birthday, 2.3 million of them in their first month of life. Nigeria accounted for about **856,000** of those deaths, around 18% of the global total and more than any other country.

Nigeria's under-5 mortality rate fell from 208 deaths per 1,000 live births in 1990 to 116 in 2024, but it has barely moved since 2015 (118). Of 200 countries, 66 are still above the SDG under-5 target, 74 above the newborn target, and 78 of 195 above the maternal mortality benchmark.

These figures come from the downloaded UN IGME and UN MMEIG estimates (see section 5) and will be re-checked in the final dashboard.

3. Questions the dashboard will answer

1. **Where is the world now?** How have under-5, newborn and maternal death rates changed since 1990 and 2000, globally and by region and income group?
2. **Who is on track for 2030?** Which countries have already met each target, which are on track at their current pace, and which need to speed up?
3. **How much faster?** For each off-track country, what annual rate of reduction has it achieved since 2015, and what rate would it need to reach the target by 2030?
4. **What would it take for Nigeria?** If Nigeria improved at a chosen pace (for example, the pace of a peer country), when would it reach each target, and how many children's lives would that represent?
5. **How sure are we?** How wide are the uncertainty ranges around the estimates, and where is the evidence thinnest?

4. Indicators and targets

Indicator	Definition	SDG target	Coverage in data
Under-5 mortality rate	Deaths before age 5 per 1,000 live births	≤ 25 by 2030 (SDG 3.2)	1990–2024, with uncertainty range
Neonatal mortality rate	Deaths in the first 28 days per 1,000 live births	≤ 12 by 2030 (SDG 3.2)	1990–2024, with uncertainty range
Maternal mortality ratio	Maternal deaths per 100,000 live births	Global < 70 by 2030 (SDG 3.1); no country above 140	2000–2023, point estimate only
Infant mortality rate	Deaths before age 1 per 1,000 live births	Context (no SDG target)	1990–2024
Stillbirth rate	Stillbirths per 1,000 total births	Context (Every Newborn target ≤ 12 by 2030)	2000–2023
Under-5 and neonatal deaths	Number of deaths	Context	1990–2024
Births and population	UN World Population Prospects	Used for weighting	1990–2023

SDG 3.1 sets a global target for maternal mortality. For individual countries the dashboard will use 70 as the benchmark and also flag the supplementary SDG aim that no country should be above 140.

5. Data sources

Source	What it provides	File
UN Inter-agency Group for Child Mortality Estimation (UN IGME), via the UNICEF Data Warehouse	Under-5, infant and neonatal mortality rates and deaths, stillbirth rate, with lower and upper bounds	data/raw/unicef_mortality.csv
UN Maternal Mortality Estimation Inter-agency Group (MMEIG) 2025, via the UNICEF Data Warehouse	Maternal mortality ratio, 2000–2023	same file
UN DESA World Population Prospects, via the UNICEF Data Warehouse	Births and total population	same file
World Bank country and lending groups	Region and income group for each country	data/raw/CLASS.xlsx

The mortality file holds 172,654 rows for 200 countries and about 200 regional groupings. All figures are modelled estimates, not direct counts.

6. Methods

Annual rate of reduction (ARR)

The standard UN IGME measure of progress: the average yearly percentage decline in a rate between two years.

```
ARR = ln(rate in year 1 / rate in year 2) / (year 2 – year 1)
```

Achieved ARR is measured from 2015, when the SDGs began, to the latest year. Required ARR is the rate needed from the latest year to reach the target in 2030:

```
Required ARR = ln(latest rate / target) / (2030 – latest year)
```

Progress status

- **Target met:** latest rate is at or below the target.
- **On track:** achieved ARR is at least the required ARR.
- **Acceleration needed:** improving, but slower than required.
- **No progress:** the rate has not fallen since 2015.

Projections and what-if

A country's rate in 2030 at its current pace, and the year it would reach the target, follow from the ARR:

```
Projected rate in 2030 = latest rate × e^(–ARR × (2030 – latest year))
Year target reached = latest year + ln(latest rate / target) / ARR
```

A what-if slider lets the user set a different ARR for Nigeria (or any country) and see the effect. These are simple trend extrapolations for illustration, not forecasts.

Aggregates

Regional, income group and world figures use the official UN IGME and MMEIG aggregates included in the download, which are weighted by births. The dashboard does not average country rates, because that would give a small country the same weight as Nigeria. The World Bank regional rows in the download carry no mortality figures, so regional trends use UNICEF reporting regions; each country's own region and income group come from the World Bank FY27 groups.

7. Dashboard pages

Page	Content
1. Global picture	Headline numbers (deaths, rates, Nigeria's share); trends since 1990 by region; rates by income group; countries by status; indicator switch.
2. On track for 2030?	Achieved vs required ARR for every off-track country; ranking switchable between seven measures; region filter.
3. Compare countries	Every country's rate in 1990, 2015 and the latest year with change and yearly reduction, beside its status and outlook.
4. Country profile (drill-through)	Right-click any country to open: trends with uncertainty bands, the target line and progress on all three targets.
5. Nigeria in focus	Stalled progress since 2015; projections at current, chosen (slider), peer-country and required pace; lives saved by 2030.
6. About the data	Sources, definitions, methods, uncertainty and limitations.
Hidden tooltip page	Status and mini trend for any country on hover.

8. Power BI techniques

Techniques chosen because they serve the questions above, not for show:

- **Field parameter** to switch the country ranking between seven measures (latest rate, change, yearly reduction achieved or needed, speed-up needed, gap and deaths above target). A slicer switches indicators.
- **Calculation group** for latest value, 2015 baseline, change since 2015 and ARR, so each logic is written once.
- **What-if parameter** for the rate of reduction in the Nigeria scenario.
- **Drill-through** from any chart or table to the country profile.
- **Bookmarks** for a guided story through the main findings.
- **Uncertainty bands** (error bars) around child mortality trends.
- **Small multiples,** dynamic titles, conditional formatting with status icons and a report-page tooltip.
- **Project format (PBIP)** so the model and pages are text files that can be reviewed in Git, with the cleaning and page generation scripted in Python.

9. Deliverables

- Power BI project (.pbip) with six pages, a tooltip page, five bookmarks and a custom theme.
- Python cleaning script producing tidy tables (facts, countries, indicators, years, targets).
- README with screenshots, findings, methods and limitations.
- Public GitHub repository and a short LinkedIn post.

10. Limitations

- All figures are modelled estimates. For countries with weak death registration, including Nigeria, they rest on surveys and models and carry wide uncertainty.
- Maternal mortality has no uncertainty range in this download and ends in 2023; child mortality ends in 2024.
- Births and population end in 2023, so 2024 rates rely on the UN IGME figures themselves.
- Projections assume a constant rate of reduction. Real progress depends on conflict, health financing, health workers and coverage of key interventions, which are outside this dataset.
- SDG 3.1 is a global target; country benchmarks of 70 and 140 are used for comparison only.

11. Plan

#	Step	Status
1	Download UN IGME, MMEIG and population data	Done
2	Download World Bank country groups (CLASS.xlsx)	Done
3	Write this project brief	Done
4	Clean and reshape data; compute ARR, status and projections; check against published figures	Done
5	Build data model and measures (field parameter, calculation group, what-if)	Done
6	Build pages, theme, drill-through, bookmarks and tooltip	Done
7	Review findings and accuracy; screenshots	Screenshots pending

#	Step	Status
8	README and GitHub	README done; GitHub pending